

1. (1 pt) The slope at a point on the velocity vs time graph is the:

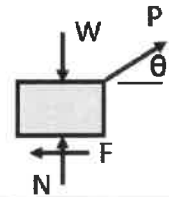
Change in position	Current position	Change in acceleration	Current acceleration
☐	☐	☐	☒

2. (1 pt) Express the vector 12 lb at
- 155°
- from CCW +x-axis in terms of unit vectors.

$10.9\hat{i} + 5.1\hat{j}$ lb	$-10.9\hat{i} + 5.1\hat{j}$ lb	$5.1\hat{i} + 10.9\hat{j}$ lb	$-5.1\hat{i} + 10.9\hat{j}$ lb
☐	☒	☐	☐

3. (1 pt) Consider a person that is standing in an elevator that is accelerating upward. How does the upward normal force exerted by the elevator floor on the person, N, compare to their weight, W?

$N > W$	$N = W$	$N < W$
☒	☐	☐

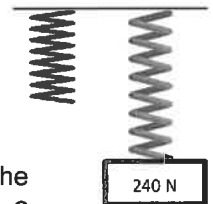


4. (1 pt) The value of the normal force is:

W	$W + P\cos\theta$	$W + P\sin\theta$	$W - P\cos\theta$	$W - P\sin\theta$
☐	☐	☐	☐	☒

5. (1 pt) A spring is hung from a ceiling as shown. When a 240 N weight is attached to the end of the spring, the spring stretches 46 cm. What is the spring constant of this spring?

11040 N/m	5.22 N/m	110 N/m	521 N/m
☐	☐	☐	☒



6. (1 pt) A ball is dropped straight down and bounces on the ground. The ball's speed before the bounce is 3 ft/s. The coefficient of restitution is 0.78. What is the ball's speed after the bounce?

2.34 ft/s	3 ft/s	3.85 ft/s
☒	☐	☐

7. (1 pt) An object is rotating with a period of 0.6 seconds. The frequency is:

0.095 rev/sec	1.67 rev/sec	3.77 rev/sec	10.47 rev/sec
☐	☒	☐	☐

8. (1 pt) The cross product of
- $\hat{k} \times \hat{i}$
- is:

0	$\hat{i}$	$-\hat{i}$	$\hat{j}$	$-\hat{j}$	$\hat{k}$	$-\hat{k}$
☐	☐	☐	☒	☐	☐	☐

9. (3 pts) An Amazon fulfillment center has a floor area of 18.5 acres. How many pallets can be laid down on the floor of this Amazon Fulfillment center?

1 acre = 43,560 ft²

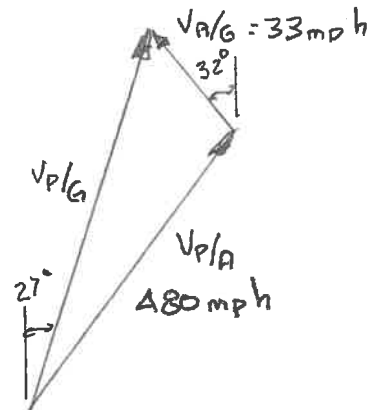
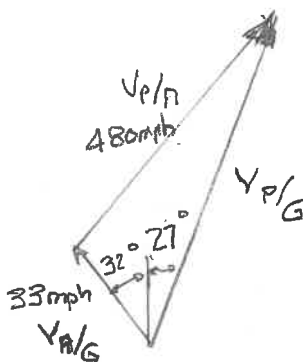
Pallet dimensions = 48 in. x 40 in.

60400 pallets

$$\text{Pallet area} = 48 \text{ in.} \times 40 \text{ in.} = 1920 \text{ in.}^2 \left(\frac{1 \text{ ft}}{12 \text{ in.}} \right)^2 = 13.33 \text{ ft}^2$$

$$\# \text{ Pallets} = 18.5 \text{ acre} \left(\frac{43560 \text{ ft}^2}{\text{acre}} \right) \left(\frac{1 \text{ pallet}}{13.33 \text{ ft}^2} \right) = 60440 \text{ pallets}$$

10. (3 pts) A plane needs to fly at 27° E of N to reach the desired destination. The windspeed is 33 mph at 32° W of N. The plane can maintain an airspeed of 480 mph. Draw a complete relative velocity vector diagram of this scenario. You must label the vectors with the following labels [$V_{p/g}$, $V_{p/a}$, $V_{a/g}$] and including all given information on your diagram. DO NOT SOLVE!



11. (3 pts) Michael drives a scooter at a constant acceleration of 3 ft/s² straight across the Perkins Quad. He starts from rest. How far has he traveled when his speed is 19 ft/s?

60.2 ft

$$V_1 = 0$$

$$V_2 = 19 \text{ ft/s}$$

$$a = 3 \text{ ft/s}^2$$

$$S_1 = 0$$

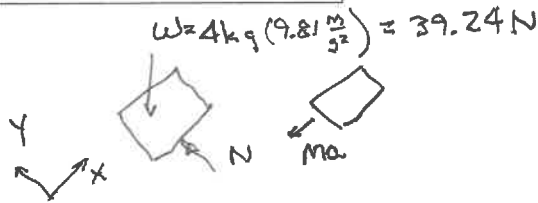
$$S_2 = ?$$

$$S_2 = S_1 + \frac{V_2^2 - V_1^2}{2a}$$

$$S_2 = 0 + \frac{(19 \text{ ft/s})^2 - (0)^2}{2(3 \text{ ft/s}^2)} = 60.17 \text{ ft}$$

12. (3 pts) Determine the acceleration of the cart.

$$4.30 \text{ m/s}^2$$

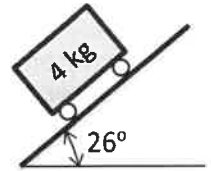


$$x: mg \sin 26^\circ = ma$$

$$g \sin 26^\circ = a$$

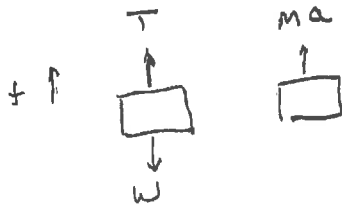
$$(9.81 \frac{\text{m}}{\text{s}^2}) \sin 26^\circ = 4.300 \text{ m/s}^2$$

$$\text{or } 14.17 \text{ ft/s}^2$$



13. (3 pts) A drone is lifting a package with an upward acceleration of 2 m/s^2 . Determine the tension in cable connected to the package.

$$35.4 \text{ N}$$

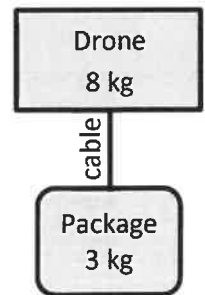


$$T - W = ma$$

$$T = W + ma$$

$$T = 3 \text{ kg} (9.81 \frac{\text{m}}{\text{s}^2}) + 3 \text{ kg} (2 \frac{\text{m}}{\text{s}^2})$$

$$= 35.43 \text{ N}$$



14. (3 pts) A 8 m radius Ferris Wheel is rotating at 2.1 revolutions per minute. Determine the normal acceleration of a 70 kg rider.

$$0.387 \frac{\text{m}}{\text{s}^2}$$

$$a_n = \omega^2 r$$

$$a_n = (0.2199 \frac{\text{rad}}{\text{s}})^2 (8 \text{ m})$$

$$= 0.3869 \frac{\text{m}}{\text{s}^2}$$

$$\omega = 2.1 \frac{\text{rev}}{\text{min}} \cdot \frac{1 \text{ min}}{60 \text{ sec}} \cdot \frac{2\pi \text{ rad}}{1 \text{ rev}} = 0.2199 \frac{\text{rad}}{\text{s}}$$

$$v = \omega r = 0.2199 \frac{\text{rad}}{\text{s}} (8 \text{ m}) = 1.759 \text{ m/s}$$

15. (3 pts) A 15 lb box is pushed across a frictionless surface with a 10 lb force (as shown). If the box starts from rest, how far has it traveled when it reaches a speed of 17 ft/s?

$$6.73 \text{ ft}$$

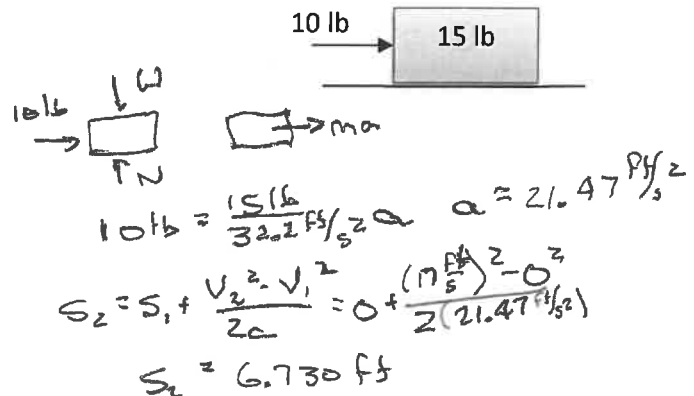
$$W = \Delta KE$$

$$F_d = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2$$

$$10 \text{ lb}(d) = \frac{1}{2} \left(\frac{15 \text{ lb}}{32.2 \text{ ft/s}^2} \right) (17 \text{ ft/s})^2$$

$$10d = 67.31$$

$$d = 6.731 \text{ ft}$$



16. (3 pts) A variable force is applied to a 7 kg cart, originally at rest, for a time, t_{final} . During this time, the cart's speed increases to 10 m/s. What is t_{final} ?

$$3.18 \text{ sec}$$

$$I_{\text{imp}} = m v_2 - m v_1$$

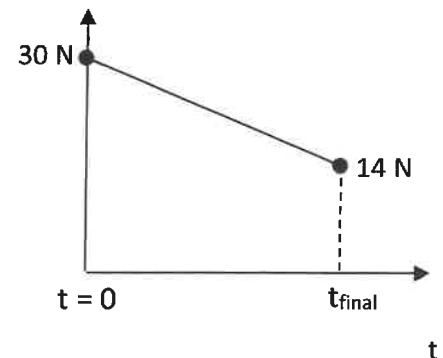
$$22 t_f = 7 \text{ kg} (10 \text{ m/s})$$

$$t_f = 3.182 \text{ sec}$$

$$I_{\text{imp}} =$$

$$\frac{1}{2} (30 \text{ N} + 14 \text{ N}) t_f$$

$$= 22 t_f$$



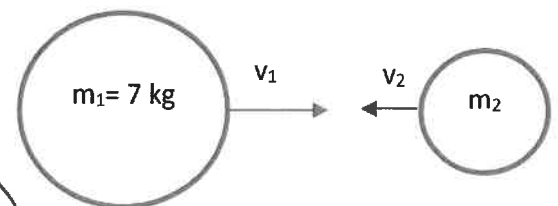
17. (3 pts) The two objects (shown) have a 1D perfectly inelastic collision. Before the collision, m_1 has a velocity of +4 m/s and m_2 has a velocity of -6 m/s. After the collision, the two objects have a speed of +2 m/s. What is the mass of m_2 ?

$$1.75 \text{ kg}$$

$$m_1 v_1 + m_2 v_2 = (m_1 + m_2) v'$$

$$7 \text{ kg} (4 \text{ m/s}) + m_2 (-6 \text{ m/s}) = (7 \text{ kg} + m_2) (2 \text{ m/s})$$

$$m_2 = 1.75 \text{ kg}$$



18. (3 pts) Determine the y-coordinate of the center of mass. Assume uniform thickness and density.

$$8.0 \text{ inch}$$

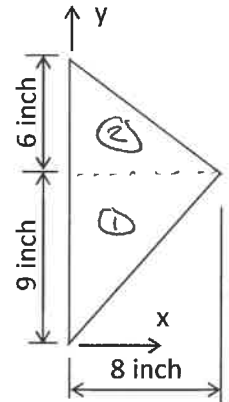
$$\bar{y} = \frac{A_1 y_1 + A_2 y_2}{A_1 + A_2}$$

$$\bar{y} = \frac{36 \text{ in}^2 (6 \text{ in}) + 24 \text{ in}^2 (9 \text{ in} + 2 \text{ in})}{60 \text{ in}^2} = 8.00 \text{ in.}$$

$$A_1 = \frac{1}{2} (9 \text{ in.}) (8 \text{ in.}) = 36 \text{ in.}^2$$

$$A_2 = \frac{1}{2} (6 \text{ in.}) (8 \text{ in.}) = 24 \text{ in.}^2$$

$$A = 36 + 24 = 60 \text{ in.}^2$$



19. (3 pts) Determine the kinetic energy of a uniform disk with a mass of 22 kg and a radius of 0.38 m that is rolling along the ground with an angular speed of 60 rad/sec.

$$8580 \text{ J}$$

$$\frac{1}{2} m (\omega^2 r^2) + \frac{1}{2} \left(\frac{1}{2} m r^2 \right) (\omega)^2$$

$$= \frac{3}{4} m \omega^2 r^2 = 8577 \text{ J}$$

$$\omega = 60 \text{ rad/s}$$

$$v = \omega r = 60 \frac{\text{rad}}{\text{s}} (0.38 \text{ m}) = 22.8 \text{ m/s}$$

$$I = \frac{1}{2} m r^2 = \frac{1}{2} (22 \text{ kg}) (0.38 \text{ m})^2 = 1.589 \text{ kg} \cdot \text{m}^2$$

$$KE = \frac{1}{2} m v^2 + \frac{1}{2} I \omega^2 = \frac{1}{2} (22 \text{ kg}) (22.8 \text{ m/s})^2 + \frac{1}{2} (1.589 \text{ kg} \cdot \text{m}^2) (60 \frac{\text{rad}}{\text{s}})^2$$

$$= 5718 \text{ J} + 2860 \text{ J} = 8578 \text{ J}$$

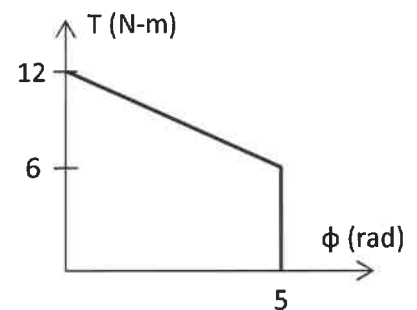
20. (3 pts) A variable torque is applied to a flywheel as shown. Determine the amount of work the torque did on the flywheel.

$$45 \text{ J}$$

$$\text{Work} = \text{Area of } \tau - \phi \text{ diagram}$$

$$W = \frac{1}{2} (12 \text{ N} \cdot \text{m} + 6 \text{ N} \cdot \text{m}) (5 \text{ rad})$$

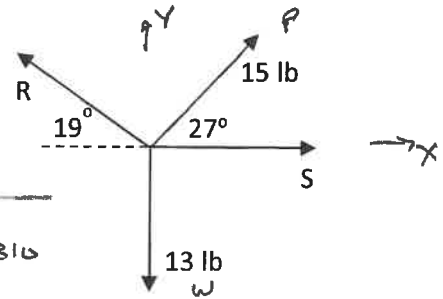
$$= 45 \text{ J}$$



21. (7 pts) The system shown at the right is at equilibrium. What is the magnitude of **S**?

4.61 lb

Force	Mag	Direction	X-comp	Y-comp
W	13 lb	↓	0	-13
P	15 lb	↗ 27°	$15 \cos 27^\circ = 13.365$	$15 \sin 27^\circ = 6.810$
S	S	→	S	0
R	R	↖ 19°	$-R \cos 19^\circ = -0.9455R$	$R \sin 19^\circ = 0.3256R$



$$X: 13.365 + S - 0.9455R = 0$$

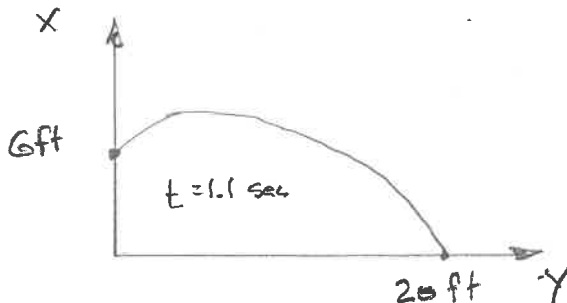
$$Y: -13 + 6.810 + 0.3256R = 0$$

$$R = 19.01 \text{ lb}$$

$$S = 4.610 \text{ lb}$$

22. (7 pts) Sydney throws a ball from a starting height of 6 ft across the room. The ball hits the floor 20 ft away in 1.1 sec. At what angle did Sydney throw the ball?

34.0°



$$X: X = X_0 + V_x t$$

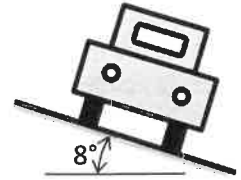
$$20 \text{ ft} = 0 + V_x (1.1 \text{ sec}) \quad V_x = 18.18 \text{ ft/s}$$

$$Y: Y = Y_0 + V_{y0} t + \frac{1}{2} a_y t^2$$

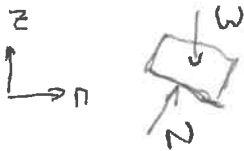
$$0 = 6 \text{ ft} + V_{y0} (1.1 \text{ sec}) + \frac{1}{2} (-32.2 \frac{\text{ft}}{\text{s}^2}) (1.1 \text{ s})^2 \quad V_{y0} = 12.26 \text{ ft/s}$$

$$\Theta = \tan^{-1} \left(\frac{V_{y0}}{V_x} \right) = \tan^{-1} \left(\frac{12.26 \text{ ft/s}}{18.18 \text{ ft/s}} \right) = 33.99^\circ$$

23. (7 pts) A car is going around a banked turn with a radius of 250 ft. Determine the required speed of the car so there is no friction. (FBD = KD Required)



$$33.6 \text{ ft/s}$$



$$z: N \cos 8^\circ - mg = 0 \Rightarrow N = \frac{mg}{\cos 8^\circ}$$

$$n: N \sin 8^\circ = ma_n \Rightarrow \frac{mg}{\cos 8^\circ} \sin 8^\circ = m \frac{v^2}{r}$$

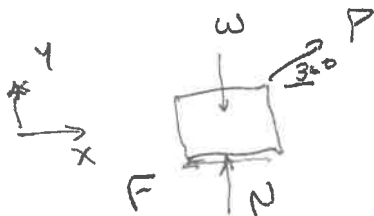
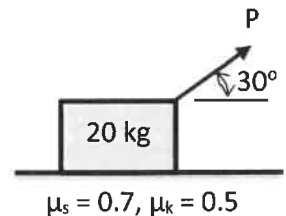
$$g \tan 8^\circ = \frac{v^2}{r}$$

$$v = \sqrt{g r \tan 8^\circ} = \sqrt{32.2 \frac{\text{ft}}{\text{s}^2} (250 \text{ ft}) (\tan 8^\circ)}$$

$$v = 33.64 \text{ ft/s}$$

24. (7 pts) Determine the minimum force P required to move the block. (FBD Required)

$$113 \text{ N}$$



$$x: P \cos 30^\circ - F = 0$$

$$y: N + P \sin 30^\circ - W = 0$$

$$\text{Impending motion: } F = \mu_s N$$

$$\Rightarrow N = P \sin 30^\circ + mg$$

$$P \cos 30^\circ - \mu_s (P \sin 30^\circ + mg) = 0$$

$$P \cos 30^\circ - 0.7 (P \sin 30^\circ + 20 \text{ kg} (9.81 \frac{\text{m}}{\text{s}^2})) = 0$$

$$P = 112.9 \text{ N}$$

25. (7 pts) A 65 kg cart with a fan starts from rest at the top of a hill. The fan is turned on, providing the cart with 1250 J of energy. When the cart reaches the bottom of the hill, it compresses a spring ($k = 3200 \text{ N/m}$) 1.1 m before coming to a stop. How much energy loss occurs during the cart's trip?

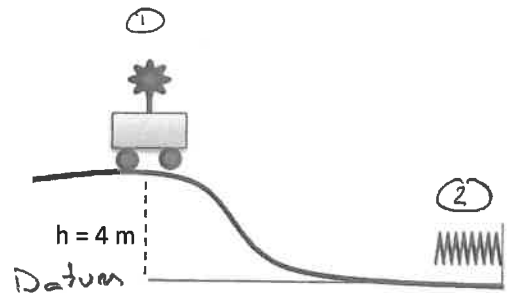
1860 J

$$mgh_1 + \frac{1}{2}mv_1^2 + \frac{1}{2}kx_1^2 + W_{\text{fan}} = mgh_2 + \frac{1}{2}mv_2^2 + \frac{1}{2}kx_2^2 + E_{\text{loss}}$$

$$65 \text{ kg} (9.81 \frac{\text{m}}{\text{s}^2}) (4 \text{ m}) + 1250 \text{ J} \\ = \frac{1}{2} (3200 \frac{\text{N}}{\text{m}}) (1.1 \text{ m})^2 + E_{\text{loss}}$$

$$2550.6 \text{ J} + 1250 \text{ J} = 1936 \text{ J} + E_{\text{loss}}$$

$$E_{\text{loss}} = 1864.6 \text{ J}$$



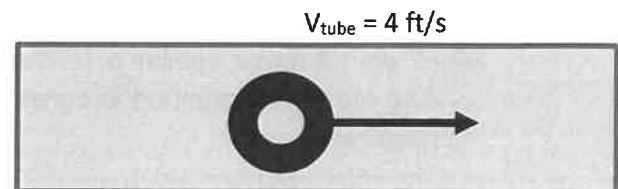
26. (7 pts) Dr. Bennett decides to go tubing after the EF 151 final. The tube (10 lb) is traveling down the river at +4 ft/s in the x-direction. Dr. Bennett (170 lb) jumps on the tube at 2 ft/s at an angle of 17° from +x-axis (CCW). What is the direction of the velocity of the tube after Dr. Bennett jumps on?

15.2°

$$x: m_b v_b \cos 17^\circ + m_t v_t = (m_b + m_t) v_x' \\ \frac{170 \text{ lb}}{g} (2 \frac{\text{ft}}{\text{s}}) \cos 17^\circ + \frac{10 \text{ lb}}{g} (4 \frac{\text{ft}}{\text{s}}) = (\frac{170+10}{g}) v_x' \\ v_x' = 2.028 \text{ ft/s}$$

$$y: m_b v_b \sin 17^\circ + m_t 0 = (m_b + m_t) v_y' \\ \frac{170 \text{ lb}}{g} (2 \frac{\text{ft}}{\text{s}}) \sin 17^\circ = \frac{180 \text{ lb}}{g} v_y' \\ v_y' = 0.552 \text{ ft/s}$$

$$\theta = \tan^{-1} \left(\frac{v_y'}{v_x'} \right) = \tan^{-1} \left(\frac{0.552 \text{ ft/s}}{2.028 \text{ ft/s}} \right) = 15.23^\circ$$



$v_{\text{bennett}} = 2 \text{ ft/s}$

27. (7 pts) A 3 kg, 0.8 m long plank is released from rest at the horizontal position. When the plank is vertical, a 0.6 kg blob of clay hits the rod and sticks to it, bringing the rod to rest. Determine the speed of the clay.

$$12.9 \text{ m/s}$$

CoE: plank dropping

$$mgh = \frac{1}{2} I \omega^2$$

$$3 \text{ kg} (9.81 \frac{\text{m}}{\text{s}^2}) (0.4 \text{ m}) = \frac{1}{2} (0.64 \text{ kg} \cdot \text{m}^2) \omega^2$$

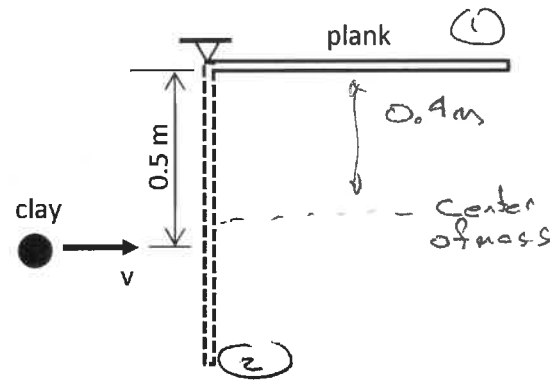
$$\omega = 6.06 \text{ rad/s}$$

COM

$$m_c v_c r_c - I_p \omega_p = 0$$

$$0.6 \text{ kg} (v_c) (0.5 \text{ m}) - 0.64 \text{ kg} \cdot \text{m}^2 (6.06 \text{ rad/s}) = 0$$

$$v_c = 12.94 \text{ m/s}$$



$$I = \frac{1}{12} m L^2 + m \left(\frac{L}{2}\right)^2$$

$$= 3 \text{ kg} (0.8 \text{ m})^2 \left\{ \frac{1}{12} + \frac{1}{4} \right\}$$

$$= 0.64 \text{ kg} \cdot \text{m}^2$$

28. (7 pts) A motor applies a 15 N-m torque to the pulley as shown to lift the 4 kg mass. Determine the acceleration of the mass. (2 FBD = KD required)

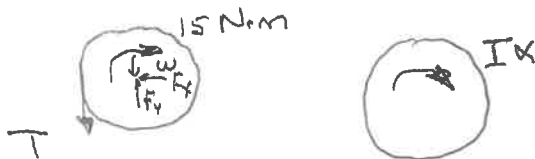
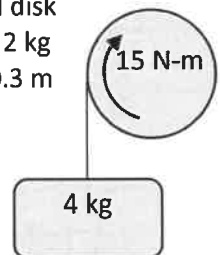
$$2.15 \text{ m/s}^2 \uparrow$$

$$I = \frac{1}{2} m r^2$$

$$= \frac{1}{2} (2 \text{ kg}) (0.3 \text{ m})^2$$

$$= 0.09 \text{ kg} \cdot \text{m}^2$$

Solid disk
m = 2 kg
r = 0.3 m



$$-15 \text{ N} \cdot \text{m} + T(r) = -I \alpha$$

$$-15 + T(0.3) = 0.09 \alpha$$

$$T - w = ma$$

$$T - 4 \text{ kg} (9.81 \text{ m/s}^2) = 4 \text{ kg} (a)$$

$$a = \alpha r = 0.3 \alpha$$

$$T = 47.85 \text{ N}$$

$$a = 2.152 \text{ m/s}^2$$

$$\alpha = 7.173 \text{ rad/s}^2$$

